

Exam: Time Series Visualization with Bokeh

Dataset intégré directement dans ce notebook — aucun fichier à télécharger !

Initialisation & Chargement du Dataset

```
import io
import pandas as pd
from bokeh.plotting import figure, show
from bokeh.io import output_notebook
from bokeh.models import (
    ColumnDataSource, HoverTool, DatetimeTickFormatter,
    NumeralTickFormatter, DateRangeSlider, CustomJS
)
from bokeh.layouts import column
from bokeh.transform import factor_cmap
from bokeh.palettes import Spectral11

output_notebook()

# — Dataset Melbourne embarqué directement (pas besoin de fichier CSV) —
_CSV = io.StringIO('Date,Daily minimum temperatures\n1981-01-01,6.8\n1981-01-02,6.7\n1981-01-03,6.8\n1981-01-04,8.8\n1981-01-05,6.7\n1981-01-06,8.8\n1981-01-07,8.8\n1981-01-08,8.8\n1981-01-09,8.8\n1981-01-10,8.8\n1981-01-11,8.8\n1981-01-12,8.8\n1981-01-13,8.8\n1981-01-14,8.8\n1981-01-15,8.8\n1981-01-16,8.8\n1981-01-17,8.8\n1981-01-18,8.8\n1981-01-19,8.8\n1981-01-20,8.8\n1981-01-21,8.8\n1981-01-22,8.8\n1981-01-23,8.8\n1981-01-24,8.8\n1981-01-25,8.8\n1981-01-26,8.8\n1981-01-27,8.8\n1981-01-28,8.8\n1981-01-29,8.8\n1981-01-30,8.8\n1981-01-31,8.8\n1981-02-01,8.8\n1981-02-02,8.8\n1981-02-03,8.8\n1981-02-04,8.8\n1981-02-05,8.8\n1981-02-06,8.8\n1981-02-07,8.8\n1981-02-08,8.8\n1981-02-09,8.8\n1981-02-10,8.8\n1981-02-11,8.8\n1981-02-12,8.8\n1981-02-13,8.8\n1981-02-14,8.8\n1981-02-15,8.8\n1981-02-16,8.8\n1981-02-17,8.8\n1981-02-18,8.8\n1981-02-19,8.8\n1981-02-20,8.8\n1981-02-21,8.8\n1981-02-22,8.8\n1981-02-23,8.8\n1981-02-24,8.8\n1981-02-25,8.8\n1981-02-26,8.8\n1981-02-27,8.8\n1981-02-28,8.8\n1981-03-01,8.8\n1981-03-02,8.8\n1981-03-03,8.8\n1981-03-04,8.8\n1981-03-05,8.8\n1981-03-06,8.8\n1981-03-07,8.8\n1981-03-08,8.8\n1981-03-09,8.8\n1981-03-10,8.8\n1981-03-11,8.8\n1981-03-12,8.8\n1981-03-13,8.8\n1981-03-14,8.8\n1981-03-15,8.8\n1981-03-16,8.8\n1981-03-17,8.8\n1981-03-18,8.8\n1981-03-19,8.8\n1981-03-20,8.8\n1981-03-21,8.8\n1981-03-22,8.8\n1981-03-23,8.8\n1981-03-24,8.8\n1981-03-25,8.8\n1981-03-26,8.8\n1981-03-27,8.8\n1981-03-28,8.8\n1981-03-29,8.8\n1981-03-30,8.8\n1981-03-31,8.8\n1981-04-01,8.8\n1981-04-02,8.8\n1981-04-03,8.8\n1981-04-04,8.8\n1981-04-05,8.8\n1981-04-06,8.8\n1981-04-07,8.8\n1981-04-08,8.8\n1981-04-09,8.8\n1981-04-10,8.8\n1981-04-11,8.8\n1981-04-12,8.8\n1981-04-13,8.8\n1981-04-14,8.8\n1981-04-15,8.8\n1981-04-16,8.8\n1981-04-17,8.8\n1981-04-18,8.8\n1981-04-19,8.8\n1981-04-20,8.8\n1981-04-21,8.8\n1981-04-22,8.8\n1981-04-23,8.8\n1981-04-24,8.8\n1981-04-25,8.8\n1981-04-26,8.8\n1981-04-27,8.8\n1981-04-28,8.8\n1981-04-29,8.8\n1981-04-30,8.8\n1981-05-01,8.8\n1981-05-02,8.8\n1981-05-03,8.8\n1981-05-04,8.8\n1981-05-05,8.8\n1981-05-06,8.8\n1981-05-07,8.8\n1981-05-08,8.8\n1981-05-09,8.8\n1981-05-10,8.8\n1981-05-11,8.8\n1981-05-12,8.8\n1981-05-13,8.8\n1981-05-14,8.8\n1981-05-15,8.8\n1981-05-16,8.8\n1981-05-17,8.8\n1981-05-18,8.8\n1981-05-19,8.8\n1981-05-20,8.8\n1981-05-21,8.8\n1981-05-22,8.8\n1981-05-23,8.8\n1981-05-24,8.8\n1981-05-25,8.8\n1981-05-26,8.8\n1981-05-27,8.8\n1981-05-28,8.8\n1981-05-29,8.8\n1981-05-30,8.8\n1981-05-31,8.8\n1981-06-01,8.8\n1981-06-02,8.8\n1981-06-03,8.8\n1981-06-04,8.8\n1981-06-05,8.8\n1981-06-06,8.8\n1981-06-07,8.8\n1981-06-08,8.8\n1981-06-09,8.8\n1981-06-10,8.8\n1981-06-11,8.8\n1981-06-12,8.8\n1981-06-13,8.8\n1981-06-14,8.8\n1981-06-15,8.8\n1981-06-16,8.8\n1981-06-17,8.8\n1981-06-18,8.8\n1981-06-19,8.8\n1981-06-20,8.8\n1981-06-21,8.8\n1981-06-22,8.8\n1981-06-23,8.8\n1981-06-24,8.8\n1981-06-25,8.8\n1981-06-26,8.8\n1981-06-27,8.8\n1981-06-28,8.8\n1981-06-29,8.8\n1981-06-30,8.8\n1981-07-01,8.8\n1981-07-02,8.8\n1981-07-03,8.8\n1981-07-04,8.8\n1981-07-05,8.8\n1981-07-06,8.8\n1981-07-07,8.8\n1981-07-08,8.8\n1981-07-09,8.8\n1981-07-10,8.8\n1981-07-11,8.8\n1981-07-12,8.8\n1981-07-13,8.8\n1981-07-14,8.8\n1981-07-15,8.8\n1981-07-16,8.8\n1981-07-17,8.8\n1981-07-18,8.8\n1981-07-19,8.8\n1981-07-20,8.8\n1981-07-21,8.8\n1981-07-22,8.8\n1981-07-23,8.8\n1981-07-24,8.8\n1981-07-25,8.8\n1981-07-26,8.8\n1981-07-27,8.8\n1981-07-28,8.8\n1981-07-29,8.8\n1981-07-30,8.8\n1981-07-31,8.8\n1981-08-01,8.8\n1981-08-02,8.8\n1981-08-03,8.8\n1981-08-04,8.8\n1981-08-05,8.8\n1981-08-06,8.8\n1981-08-07,8.8\n1981-08-08,8.8\n1981-08-09,8.8\n1981-08-10,8.8\n1981-08-11,8.8\n1981-08-12,8.8\n1981-08-13,8.8\n1981-08-14,8.8\n1981-08-15,8.8\n1981-08-16,8.8\n1981-08-17,8.8\n1981-08-18,8.8\n1981-08-19,8.8\n1981-08-20,8.8\n1981-08-21,8.8\n1981-08-22,8.8\n1981-08-23,8.8\n1981-08-24,8.8\n1981-08-25,8.8\n1981-08-26,8.8\n1981-08-27,8.8\n1981-08-28,8.8\n1981-08-29,8.8\n1981-08-30,8.8\n1981-08-31,8.8\n1981-09-01,8.8\n1981-09-02,8.8\n1981-09-03,8.8\n1981-09-04,8.8\n1981-09-05,8.8\n1981-09-06,8.8\n1981-09-07,8.8\n1981-09-08,8.8\n1981-09-09,8.8\n1981-09-10,8.8\n1981-09-11,8.8\n1981-09-12,8.8\n1981-09-13,8.8\n1981-09-14,8.8\n1981-09-15,8.8\n1981-09-16,8.8\n1981-09-17,8.8\n1981-09-18,8.8\n1981-09-19,8.8\n1981-09-20,8.8\n1981-09-21,8.8\n1981-09-22,8.8\n1981-09-23,8.8\n1981-09-24,8.8\n1981-09-25,8.8\n1981-09-26,8.8\n1981-09-27,8.8\n1981-09-28,8.8\n1981-09-29,8.8\n1981-09-30,8.8\n1981-10-01,8.8\n1981-10-02,8.8\n1981-10-03,8.8\n1981-10-04,8.8\n1981-10-05,8.8\n1981-10-06,8.8\n1981-10-07,8.8\n1981-10-08,8.8\n1981-10-09,8.8\n1981-10-10,8.8\n1981-10-11,8.8\n1981-10-12,8.8\n1981-10-13,8.8\n1981-10-14,8.8\n1981-10-15,8.8\n1981-10-16,8.8\n1981-10-17,8.8\n1981-10-18,8.8\n1981-10-19,8.8\n1981-10-20,8.8\n1981-10-21,8.8\n1981-10-22,8.8\n1981-10-23,8.8\n1981-10-24,8.8\n1981-10-25,8.8\n1981-10-26,8.8\n1981-10-27,8.8\n1981-10-28,8.8\n1981-10-29,8.8\n1981-10-30,8.8\n1981-10-31,8.8\n1981-11-01,8.8\n1981-11-02,8.8\n1981-11-03,8.8\n1981-11-04,8.8\n1981-11-05,8.8\n1981-11-06,8.8\n1981-11-07,8.8\n1981-11-08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df = pd.read_csv(_CSV)
df.columns = ['Date', 'Temperature']
df['Date'] = pd.to_datetime(df['Date'])
df['Temperature'] = pd.to_numeric(df['Temperature'].astype(str).str.replace('.', '', regex=False), errors='coerce')
df = df.dropna(subset=['Temperature']).reset_index(drop=True)

print(f"Dataset chargé : {len(df)} lignes, de {df['Date'].min().date()} à {df['Date'].max().date()}")
df.head()
```

Dataset chargé : 3650 lignes, de 1981-01-01 à 1990-12-29

	Date	Temperature	
0	1981-01-01	6.8	
1	1981-01-02	6.7	
2	1981-01-03	6.8	
3	1981-01-04	8.8	
4	1981-01-05	6.7	

Étapes suivantes : [Générer du code avec df](#) [New interactive sheet](#)

Question 1 : Graphique linéaire de série temporelle de base

```
source = ColumnDataSource(df)

hover = HoverTool(tooltips=[
    ("Date", "@Date{%F}"),
    ("Température", "@Temperature{0.0} °C"),
], formatters={"@Date": "datetime"}, mode="vline")

p1 = figure(
    title="Daily Minimum Temperatures",
    x_axis_label="Date", y_axis_label="Temperature (°C)",
    x_axis_type="datetime", width=900, height=350,
    tools="pan,wheel_zoom,reset", toolbar_location="above"
)
p1.add_tools(hover)
p1.line("Date", "Temperature", source=source, line_color="steelblue", line_width=1.2, alpha=0.8)
p1.xaxis.formatter = DatetimeTickFormatter(months="%b %Y")

show(p1)
```

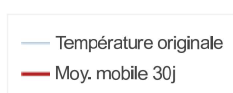
Question 2 : Moyenne mobile sur 30 jours

```
df['Rolling_Avg'] = df['Temperature'].rolling(window=30, center=True).mean()
source2 = ColumnDataSource(df)

hover2 = HoverTool(tooltips=[
    ("Date", "@Date{%F}"),
    ("Température", "@Temperature{0.0} °C"),
    ("Moy. mobile", "@Rolling_Avg{0.00} °C"),
], formatters={"@Date": "datetime"}, mode="vline")

p2 = figure(
    title="Daily Minimum Temperatures + Rolling Average (30 days)",
    x_axis_label="Date", y_axis_label="Temperature (°C)",
    x_axis_type="datetime", width=900, height=350,
    tools="pan,wheel_zoom,reset", toolbar_location="above"
)
p2.add_tools(hover2)
p2.line("Date", "Temperature", source=source2, line_color="steelblue",
        line_width=1, alpha=0.5, legend_label="Température originale")
p2.line("Date", "Rolling_Avg", source=source2, line_color="firebrick",
        line_width=2.5, line_dash="solid", legend_label="Moy. mobile 30j")
p2.legend.location = "top_left"
p2.legend.click_policy = "hide"
p2.xaxis.formatter = DatetimeTickFormatter(months="%b %Y")

show(p2)
```



Question 3 : Boîtes à moustaches mensuelles

```
import numpy as np

df['Month'] = df['Date'].dt.month
month_names = ["Jan", "Fév", "Mar", "Avr", "Mai", "Jun", "Jul", "Aoû", "Sep", "Oct", "Nov", "Déc"]

# Compute stats per month
stats = df.groupby('Month')['Temperature'].describe(percentiles=[.25, .5, .75])
stats.columns = ['count', 'mean', 'std', 'min', 'q25', 'q50', 'q75', 'max']
```

```

stats['iqr'] = stats['q75'] - stats['q25']
stats['upper'] = (stats['q75'] + 1.5 * stats['iqr']).clip(upper=stats['max'])
stats['lower'] = (stats['q25'] - 1.5 * stats['iqr']).clip(lower=stats['min'])
stats['month_name'] = month_names

src3 = ColumnDataSource(stats)
months_str = [str(m) for m in range(1,13)]

hover3 = HoverTool(tooltips=[
    ("Mois", "@month_name"),
    ("Médiane", "@q50{0.0} °C"),
    ("Min", "@min{0.0} °C"),
    ("Max", "@max{0.0} °C"),
])

p3 = figure(
    title="Distribution mensuelle des températures minimales",
    x_range=months_str, y_axis_label="Temperature (°C)",
    width=900, height=400,
    tools="pan,wheel_zoom,reset,save", toolbar_location="above"
)
p3.add_tools(hover3)

# Whiskers
p3.segment("month_name", "upper", "month_name", "q75", source=src3, line_color="black")
p3.segment("month_name", "lower", "month_name", "q25", source=src3, line_color="black")

# Boxes
p3.vbar("month_name", 0.7, "q75", "q50", source=src3, fill_color="skyblue", line_color="black")
p3.vbar("month_name", 0.7, "q50", "q25", source=src3, fill_color="lightcoral", line_color="black")

# Whisker caps
p3.rect("month_name", "upper", 0.3, 0.01, source=src3, line_color="black")
p3.rect("month_name", "lower", 0.3, 0.01, source=src3, line_color="black")

p3.xaxis.major_label_overrides = {str(i+1): month_names[i] for i in range(12)}
p3.xaxis.axis_label = "Mois"

show(p3)

```

Question 4 : Box plots annuels avec color mapping

```

df['Year'] = df['Date'].dt.year
years = sorted(df['Year'].unique())
years_str = [str(y) for y in years]

ystats = df.groupby('Year')['Temperature'].describe(percentiles=[.25,.5,.75])
ystats.columns = ['count', 'mean', 'std', 'min', 'q25', 'q50', 'q75', 'max']
ystats['iqr'] = ystats['q75'] - ystats['q25']
ystats['upper'] = (ystats['q75'] + 1.5*ystats['iqr']).clip(upper=ystats['max'])
ystats['lower'] = (ystats['q25'] - 1.5*ystats['iqr']).clip(lower=ystats['min'])
ystats['year_str'] = years_str
ystats = ystats.reset_index()

src4 = ColumnDataSource(ystats)

palette = Spectral11[:len(years)]
cmap = factor_cmap('year_str', palette=palette, factors=years_str)

```

```

hover4 = HoverTool(tooltips=[
    ("Année", "@year_str"),
    ("Médiane", "@q50{0.0} °C"),
    ("Min", "@min{0.0} °C"),
    ("Max", "@max{0.0} °C"),
    ("Moy.", "@mean{0.0} °C"),
])

p4 = figure(
    title="Distribution annuelle des températures minimales",
    x_range=years_str, y_axis_label="Temperature (°C)",
    width=900, height=400,
    tools="pan,wheel_zoom,reset", toolbar_location="above"
)
p4.add_tools(hover4)

p4.segment("year_str", "upper", "year_str", "q75", source=src4, line_color="gray")
p4.segment("year_str", "lower", "year_str", "q25", source=src4, line_color="gray")
p4.vbar("year_str", 0.7, "q75", "q50", source=src4, fill_color=cmap, line_color="black")
p4.vbar("year_str", 0.7, "q50", "q25", source=src4, fill_color=cmap, line_color="black", fill_alpha=0.6)
p4.rect("year_str", "upper", 0.4, 0.01, source=src4, line_color="gray")
p4.rect("year_str", "lower", 0.4, 0.01, source=src4, line_color="gray")
p4.xaxis.axis_label = "Année"

show(p4)

```

Question 5 : Sélection interactive de la plage temporelle

```

from bokeh.models import DateRangeSlider, CustomJS

src5_full = ColumnDataSource(df)
src5_view = ColumnDataSource(df)

hover5 = HoverTool(tooltips=[
    ("Date", "@Date{%F}"),
    ("Temp", "@Temperature{0.0} °C"),
], formatters={"@Date": "datetime"}, mode="vline")

p5 = figure(
    title="Températures minimales – plage sélectionnable",
    x_axis_label="Date", y_axis_label="Temperature (°C)",
    x_axis_type="datetime", width=900, height=350,
    tools="pan,wheel_zoom,reset", toolbar_location="above"
)
p5.add_tools(hover5)
p5.line("Date", "Temperature", source=src5_view, line_color="steelblue", line_width=1.2)
p5.xaxis.formatter = DatetimeTickFormatter(months="%b %Y")

slider = DateRangeSlider(
    title="Plage de dates",
    start=df['Date'].min(),
    end=df['Date'].max(),
    value=(df['Date'].min(), df['Date'].max()),
    step=1,
    width=900
)

```

```

callback = CustomJS(args=dict(src_full=src5_full, src_view=src5_view, slider=slider), code="""
    const [start, end] = slider.value;
    const full = src_full.data;
    const dates = full['Date'];
    const temps = full['Temperature'];

    const new_dates = [];
    const new_temps = [];
    for (let i = 0; i < dates.length; i++) {
        if (dates[i] >= start && dates[i] <= end) {
            new_dates.push(dates[i]);
            new_temps.push(temps[i]);
        }
    }
    src_view.data = {Date: new_dates, Temperature: new_temps};
    src_view.change.emit();
""")
slider.js_on_change('value', callback)

show(column(slider, p5))

```

Plage de dates: 01 Jan 1981 .. 29 Dec 1990



Question 6 : Décomposition de série temporelle

```

from bokeh.layouts import column as bk_column
# from bokeh.models import LinkedDataSource ← ligne supprimée

# Rééchantillonnage mensuel
df_monthly = df.set_index('Date').resample('MS')['Temperature'].mean().reset_index()
df_monthly.columns = ['Date', 'Monthly_Avg']

# Tendence : moyenne mobile 12 mois
df_monthly['Trend'] = df_monthly['Monthly_Avg'].rolling(window=12, center=True).mean()

# Saisonnalité
df_monthly['Seasonality'] = df_monthly['Monthly_Avg'] - df_monthly['Trend']

src6 = ColumnDataSource(df_monthly)

kwargs = dict(x_axis_type="datetime", width=900, height=220,
              tools="pan,wheel_zoom,reset", toolbar_location="above")

# Plot 1 : données mensuelles
p6a = figure(title="Données mensuelles", x_axis_label="", y_axis_label="Temp (°C)", **kwargs)
p6a.line("Date", "Monthly_Avg", source=src6, color="steelblue", line_width=2)

# Plot 2 : tendance
p6b = figure(title="Tendance (moy. mobile 12 mois)", x_range=p6a.x_range,
             x_axis_label="", y_axis_label="Temp (°C)", **kwargs)
p6b.line("Date", "Trend", source=src6, color="firebrick", line_width=2)

# Plot 3 : saisonnalité
p6c = figure(title="Saisonnalité", x_range=p6a.x_range,
             x_axis_label="Date", y_axis_label="Delta (°C)", **kwargs)
p6c.line("Date", "Seasonality", source=src6, color="seagreen", line_width=2)
p6c.line([df_monthly["Date"].min(), df_monthly["Date"].max()], [0,0],
         color="black", line_dash="dashed", line_width=1)

for p in [p6a, p6b, p6c]:

```

```
p.xaxis.formatter = DatetimeTickFormatter(months="%b %Y")
```

```
show(bk_column(p6a, p6b, p6c))
```

